

MINISTRY OF EARTH SCIENCES (MoES) — GOVERNMENT OF INDIA

NATIONAL INSTITUTE OF OCEAN TECHNOLOGY (NIOT, CHENNAI)
OPERATION NET-ZERO: TACTICAL RECOVERY DISPATCH ORDER

[DEMO MODE ACTIVE] This mission report contains simulated / demo acoustic survey data. Not for live maritime navigation.

Incident Ref ID:	INC-PIPELINE-003	Issue Timestamp:	2026-09-26 18:40:00 UTC
Dataset / Source:	SubPipe (CC-BY-4.0)	AUV Asset:	NIOT Autonomous Surveyor-II

1. ACOUSTIC ANOMALY SPECIFICATION & MEASUREMENT PROVENANCE

Parameter	Value / Reading	Scientific Provenance
Anomaly Class:	SUBMARINE_PIPELINE-CLASS CONTACT	MEASURED (ONNX Tiled)
AI Confidence:	64.5%	MEASURED (Class Prob)
Target Elevation:	1.25 m	DERIVED (Trigonometry)
Seafloor Depth:	24.5 m	ASSUMED (Altimeter)
Coordinates:	GPS unavailable — image-space tracking only	UNAVAILABLE

2. MULTI-PING TEMPORAL TRACKING & CONTACT PERSISTENCE

Track Identifier:	TRK-PIPE-03	Persistence Status:	NEW_CONTACT
Observations:	1 consecutive pings	Track Age:	0.0 seconds elapsed

3. SYSTEM-1 EDGE REFLEX TELEMETRY (CONVAIINNOVATIONS/LAYA)

Reflex Decision:	PASSIVE_LOG
Hazard Severity Score:	5.4 / 10.0 (Deterministic Guardrail)
Execution Latency:	28.4 ms (Sub-35ms Non-Autoregressive)
Safety Disclaimer:	Advisory only — no direct vehicle control without operator authorization.

4. SYSTEM-2 TACTICAL REASONING BRIEFING (qwen/qwen3.8-27b (Groq Cloud) — ACTIVE)

Incident Summary:	Linear benthic conduit detected across survey transect. Potential subsea utility infrastructure.
Recovery Priority:	MODERATE
Operator Directive:	Maintain pipeline perpendicular survey track; continue passive logging.
Sensor Uncertainties:	- Burial depth along eastern boundary unconfirmed - Cathodic protection condition unmeasured

Generated By:
SAMUDRA-AI Autonomous Core
NIOT Acoustic Intelligence Unit

Verified By / Watch Officer:
Commander, Coast Guard Station Mandapam
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